Year 7

REVISION PACK

Unit 1

ANSWERS

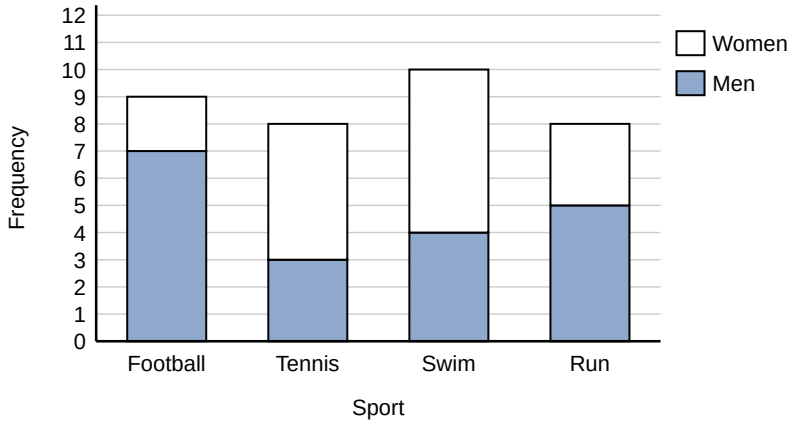
Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

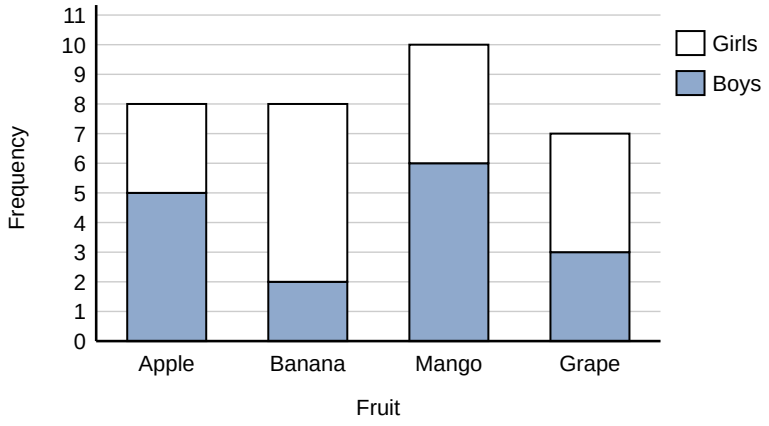
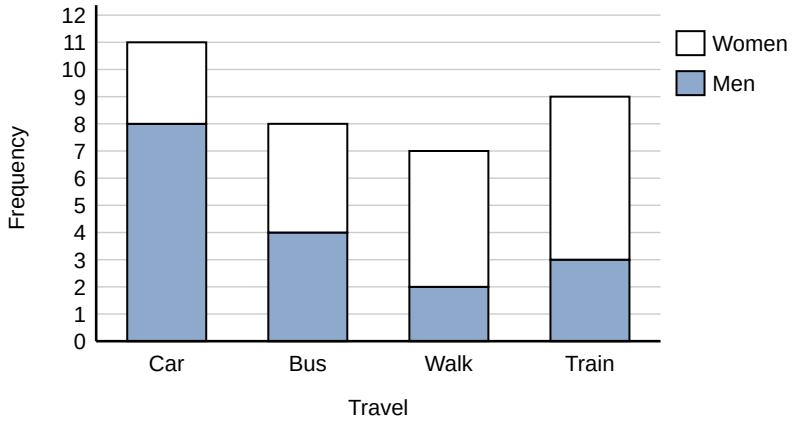
Unit 1 · Averages and range

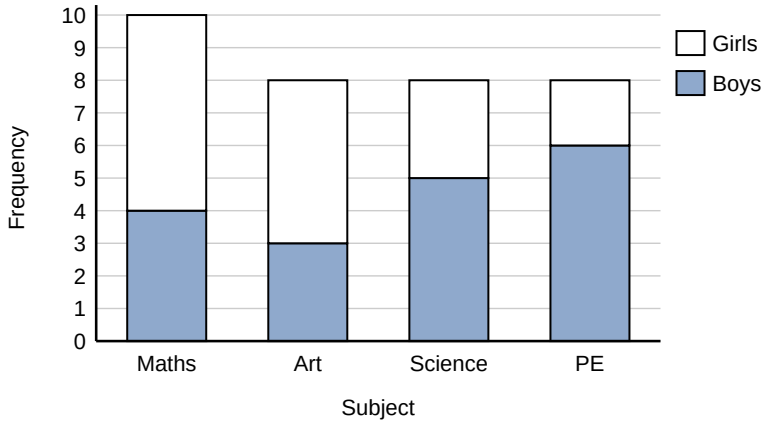
Q	Working	Answer
mode, median, mean and range		
1.1	**(a)** 3 appears the most **(b)** order: $0, 1, 2, 3, 3, 3, 5, 7$ $(3 + 3) \div 2 = 3$ **(c)** $2 + 0 + 3 + 1 + 3 + 5 + 3 + 7 = 24$ $24 \div 8 = 3$ **(d)** $7 - 0 = 7$	**(a) 3** **(b) 3** **(c) 3** **(d) 7**
1.2	**(a)** 4 appears the most **(b)** order: $0, 1, 2, 3, 4, 4, 4, 6$ $(3 + 4) \div 2 = 3.5$ **(c)** $1 + 4 + 2 + 4 + 0 + 4 + 3 + 6 = 24$ $24 \div 8 = 3$ **(d)** $6 - 0 = 6$	**(a) 4** **(b) 3.5** **(c) 3** **(d) 6**
1.3	**(a)** 1 appears the most **(b)** order: $0, 1, 1, 1, 3, 4, 6, 8$ $(1 + 3) \div 2 = 2$ **(c)** $3 + 1 + 6 + 1 + 0 + 1 + 4 + 8 = 24$ $24 \div 8 = 3$ **(d)** $8 - 0 = 8$	**(a) 1** **(b) 2** **(c) 3** **(d) 8**
1.4	**(a)** 12 appears the most **(b)** order: $5, 7, 9, 9, 12, 12, 12, 14$ $(9 + 12) \div 2 = 10.5$ **(c)** $12 + 7 + 9 + 12 + 5 + 14 + 12 + 9 = 80$ $80 \div 8 = 10$ **(d)** $14 - 5 = 9$	**(a) 12** **(b) 10.5** **(c) 10** **(d) 9**

1.5	**(a)** 25 appears the most **(b)** order: $18, 19, 20, 22, 25, 25, 25, 30$ $(22 + 25) \div 2 = 23.5$ **(c)** $20 + 25 + 18 + 25 + 30 + 22 + 25 + 19 = 184$ $184 \div 8 = 23$ **(d)** $30 - 18 = 12$	**(a) 25** **(b) 23.5** **(c) 23** **(d) 12**
1.6	**(a)** 2 appears the most **(b)** order: $1, 2, 2, 2, 3, 4, 5, 5$ $(2 + 3) \div 2 = 2.5$ **(c)** $2 + 5 + 3 + 2 + 1 + 4 + 2 + 5 = 24$ $24 \div 8 = 3$ **(d)** $5 - 1 = 4$	**(a) 2** **(b) 2.5** **(c) 3** **(d) 4**

Unit 1 · Compound bar charts

Q	Working	Answer
completing and reading a compound bar chart		
2.1	**(a)** Run: men bar to 5, women on top to 8 **(b)** $10 - 4 = 6$ **(c)** $7 + 3 + 4 + 5 = 19$ 	**(a) Run bar: men up to 5, women from 5 up to 8** **(b) 6** **(c) 19**

2.2	**(a)** Grape: boys bar to 3, girls on top to 7 **(b)** $8 - 2 = 6$ **(c)** $5 + 2 + 6 + 3 = 16$ 	**(a) Grape bar: boys up to 3, girls from 3 up to 7** **(b) 6** **(c) 16**
2.3	**(a)** Train: men bar to 3, women on top to 9 **(b)** $11 - 8 = 3$ **(c)** $8 + 4 + 2 + 3 = 17$ 	**(a) Train bar: men up to 3, women from 3 up to 9** **(b) 3** **(c) 17**

2.4	**(a)** PE: boys bar to 6, girls on top to 8 **(b)** $8 - 3 = 5$ **(c)** $4 + 3 + 5 + 6 = 18$ 	**(a) PE bar: boys up to 6, girls from 6 up to 8** **(b) 5** **(c) 18**

Unit 1 · The mean

Q	Working	Answer
the mean and adding a new value		
3.1	**(a)** $3 + 1 + 5 + 2 + 4 = 15$ $15 \div 5 = 3$ **(b)**	**(a) 3** **(b) Less: 0 is less than the mean of 3.**
3.2	**(a)** $2 + 4 + 1 + 3 + 5 = 15$ $15 \div 5 = 3$ **(b)**	**(a) 3** **(b) Greater: 6 is more than the mean of 3.**
3.3	**(a)** $6 + 9 + 4 + 7 + 9 = 35$ $35 \div 5 = 7$ **(b)**	**(a) 7** **(b) Less: 1 is less than the mean of 7.**
3.4	**(a)** $12 + 15 + 10 + 14 + 9 = 60$ $60 \div 5 = 12$ **(b)**	**(a) 12** **(b) Greater: 20 is more than the mean of 12.**
3.5	**(a)** $5 + 7 + 4 + 6 + 8 = 30$ $30 \div 5 = 6$ **(b)**	**(a) 6** **(b) Less: 2 is less than the mean of 6.**

3.6	**(a)** $4 + 0 + 6 + 3 + 2 = 15$ $15 \div 5 = 3$ **(b)**	**(a) 3** **(b) Greater: 9 is more than the mean of 3.**

Unit 1 · Dual bar charts

Q	Working	Answer
reading and comparing a dual bar chart		
4.1	**(a)** differences: $5, 2, 2, 3, 4$ **(b)** $Made: 10 + 7 + 12 + 5 + 8 = 42$ $Received: 5 + 9 + 10 + 8 + 4 = 36$	**(a) Monday** **(b) Yes: made 42, received 36**
4.2	**(a)** differences: $3, 1, 4, 6, 3$ **(b)** $Sent: 4 + 9 + 6 + 11 + 3 = 33$ $Received: 7 + 8 + 10 + 5 + 6 = 36$	**(a) Thursday** **(b) No: sent 33, received 36**
4.3	**(a)** differences: $1, 4, 4, 3, 7$ **(b)** $Sent: 5 + 3 + 8 + 6 + 2 = 24$ $Received: 6 + 7 + 4 + 3 + 9 = 29$	**(a) Friday** **(b) Yes: sent 24, received 29**
4.4	**(a)** differences: $2, 2, 6, 2, 2$ **(b)** $Made: 12 + 6 + 9 + 4 + 7 = 38$ $Received: 10 + 8 + 3 + 6 + 5 = 32$	**(a) Wednesday** **(b) Yes: made 38, received 32**

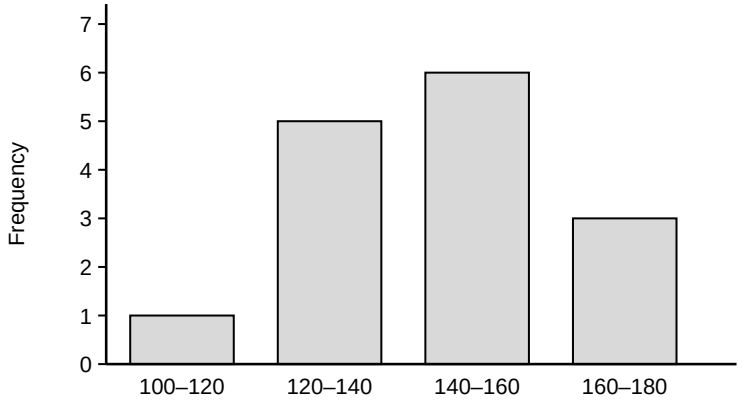
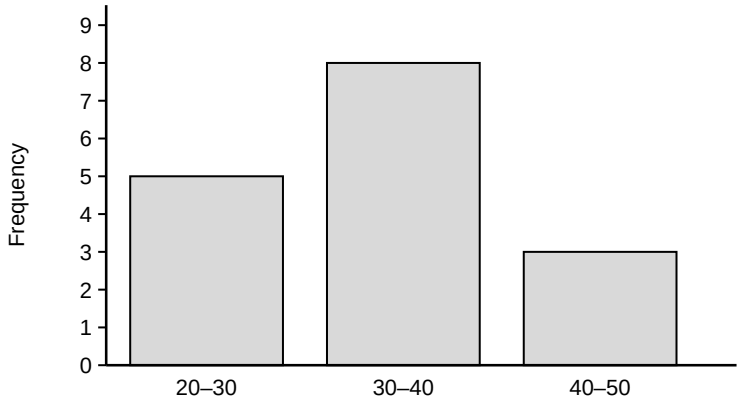
Unit 1 · Comparing data

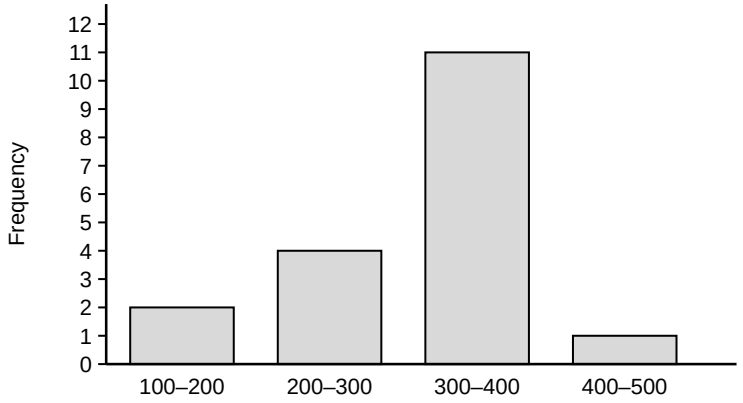
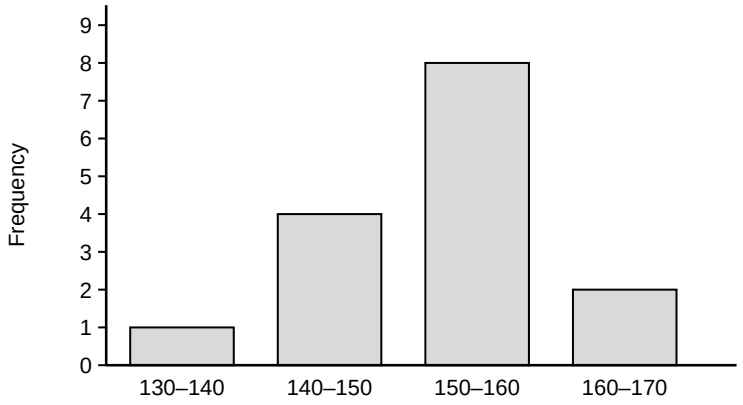
Q	Working	Answer
comparing with the median and range		
5.1	**(a)** order: $2, 4, 5, 6, 6, 8, 9$ $\text{median} = 6$ $\text{range} = 9 - 2 = 7$ **(b)**	**(a) median 6, range 7** **(b) Aziz has the higher median, so Aziz did better on average. Kareem has the smaller range, so Kareem is more consistent.**

5.2	**(a)** order: $11, 13, 14, 16, 17, 18, 20$ median $= 16$ range $= 20 - 11 = 9$ **(b)**	**(a) median 16, range 9** **(b) Hana has the higher median, so Hana did better on average. Rania has the smaller range, so Rania is more consistent.**
5.3	**(a)** order: $1, 2, 3, 4, 5, 6, 8$ median $= 4$ range $= 8 - 1 = 7$ **(b)**	**(a) median 4, range 7** **(b) Faris has the higher median, so Faris did better on average. Faris has the smaller range, so Faris is more consistent.**
5.4	**(a)** order: $25, 27, 28, 30, 31, 34, 40$ median $= 30$ range $= 40 - 25 = 15$ **(b)**	**(a) median 30, range 15** **(b) Dana has the higher median, so Dana did better on average. Dana has the smaller range, so Dana is more consistent.**
5.5	**(a)** order: $7, 9, 10, 11, 12, 13, 15$ median $= 11$ range $= 15 - 7 = 8$ **(b)**	**(a) median 11, range 8** **(b) Ibrahim has the higher median, so Ibrahim did better on average. Khalid has the smaller range, so Khalid is more consistent.**
5.6	**(a)** order: $4, 5, 6, 7, 8, 9, 10$ median $= 7$ range $= 10 - 4 = 6$ **(b)**	**(a) median 7, range 6** **(b) Salma has the higher median, so Salma did better on average. Nadia has the smaller range, so Nadia is more consistent.**

Unit 1 · Grouped data

Q	Working	Answer
grouped frequency tables		

6.1	**(a)** frequencies: $1, 5, 6, 3$ **(b)** **(c)** **(d)** $6 + 3 = 9$ 	**(a) 100 < M ≤ 120: 1, 120 < M ≤ 140: 5, 140 < M ≤ 160: 6, 160 < M ≤ 180: 3** **(b) Bars with heights 1, 5, 6, 3 (no gaps)** **(c) 140 < M ≤ 160** **(d) 9**
6.2	**(a)** frequencies: $5, 8, 3$ **(b)** **(c)** **(d)** $8 + 3 = 11$ 	**(a) 20 < T ≤ 30: 5, 30 < T ≤ 40: 8, 40 < T ≤ 50: 3** **(b) Bars with heights 5, 8, 3 (no gaps)** **(c) 30 < T ≤ 40** **(d) 11**

6.3	**(a)** frequencies: $2, 4, 11, 1$ **(b)** **(c)** **(d)** $11 + 1 = 12$ 	**(a) 100 < M ≤ 200: 2, 200 < M ≤ 300: 4, 300 < M ≤ 400: 11, 400 < M ≤ 500: 1** **(b) Bars with heights 2, 4, 11, 1 (no gaps)** **(c) 300 < M ≤ 400** **(d) 12**
6.4	**(a)** frequencies: $1, 4, 8, 2$ **(b)** **(c)** **(d)** $8 + 2 = 10$ 	**(a) 130 < H ≤ 140: 1, 140 < H ≤ 150: 4, 150 < H ≤ 160: 8, 160 < H ≤ 170: 2** **(b) Bars with heights 1, 4, 8, 2 (no gaps)** **(c) 150 < H ≤ 160** **(d) 10**

Unit 1 · Line graphs

Q	Working	Answer
reading and comparing line graphs		
7.1	**(a)** at 0 minutes: 20 **(b)** $LiquidA$: 4 min, $LiquidB$: 6 min **(c)** $90 - 70 = 20$	**(a) 20 °C** **(b) Liquid A reached 100 °C 2 minutes sooner (Liquid A 4 min, Liquid B 6 min).** **(c) 20 °C**

7.2	**(a)** at 0 minutes: 10 **(b)** $LiquidA$: 3 min, $LiquidB$: 4 min **(c)** $40 - 40 = 0$	**(a) 10 °C** **(b) Liquid A reached 60 °C 1 minutes sooner (Liquid A 3 min, Liquid B 4 min).** **(c) 0 °C**
7.3	**(a)** at 0 minutes: 90 **(b)** $LiquidA$: 2 min, $LiquidB$: 4 min **(c)** $50 - 30 = 20$	**(a) 90 °C** **(b) Liquid A reached 50 °C 2 minutes sooner (Liquid A 2 min, Liquid B 4 min).** **(c) 20 °C**
7.4	**(a)** at 0 minutes: 30 **(b)** $LiquidA$: 2 min, $LiquidB$: 3 min **(c)** $80 - 65 = 15$	**(a) 30 °C** **(b) Liquid A reached 60 °C 1 minutes sooner (Liquid A 2 min, Liquid B 3 min).** **(c) 15 °C**